RazorCAD User Manual

Parametric parts, assemblies, sheet metal, structures and drawings

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1 Introduction

1.1 What RazorCAD is

RazorCAD is a parametric part modeller for Windows with a classic feature-tree interface. You build a part as a **feature history**: sketches, extrudes, revolves, cuts, fillets, patterns and so on, each of which can be edited later. Changing a dimension or an equation rebuilds every feature after it.

Solid modelling runs on the **OpenCascade** B-rep kernel (the engine behind FreeCAD), through the replicad library. Display uses three.js. RazorCAD runs fully offline.

Main capabilities:

- Constraint-driven 2D sketches with relations, dimensions and equations, plus gear and airfoil generators
- 3D sketches for frames, pipe and duct routes and sweep paths
- Solid features: extrude, revolve, sweep, loft, hole, helix, threads, wrap, draft, rib, dome, patterns, mirror, fillet, chamfer, shell
- Multi-body parts with combine, subtract, split, move/copy, per-body materials and appearances
- Surface modelling: extruded, revolved, planar, lofted, swept, filled, boundary and ruled surfaces, trim, extend, knit, thicken and more
- Sheet metal: flanges, hems, miter flanges, sketched bends, jogs, forming tools, tab and slot, lofted bends, transitions, flat patterns and DXF export
- **Fast Structure** (weldments): structural members from standard profiles, end caps, gussets, weld beads and a cut list
- Assemblies with mates, sub-assemblies, configurations, exploded views and a bill of materials
- 2D drawings of parts and assemblies, with projected, section, detail, auxiliary, broken and flat pattern views, dimensions, GD&T and PDF and DXF export
- STEP, STL, OBJ and 3MF import and export; DXF import into sketches and DXF export of faces and flat patterns
- An **Ask Claude** panel that builds features from a plain-language description

1.2 About this manual

This manual describes RazorCAD as it is in the version shown on the cover. It is checked against the software every day and updated when features change; Appendix C lists what changed in each revision.

The **Contents** and the **Index** are clickable. The footer of every page has a **Back to Contents** link. Your PDF viewer's bookmarks pane shows the same outline.

1.3 Conventions

Convention	Meaning
File ▸ Save	A menu command: the File menu, then Save
`Ctrl+S`	A key or key combination
Property panel	The panel on the left that holds a feature's settings while you edit it
Feature tree	The feature history list on the left

Convention	Meaning
Command ribbon	The tabbed toolbar across the top (Features, Sketch, Surfaces …)
`width/2`	Text you type, such as a value or an equation

> **Note:** RazorCAD works in **millimetres**. Angles are in **degrees**.

2 Installing and Starting RazorCAD

2.1 Running RazorCAD

Double-click **RazorCAD.exe**. It is one standalone program: it needs no browser, Python or installation, and it runs offline. It uses the Microsoft Edge WebView2 runtime, which comes with Windows 10 and 11.

2.2 Installing with the setup program

Testers can be given **RazorCAD-Setup.exe**, a single setup program with RazorCAD inside it.

- Install for the current user (`%LOCALAPPDATA%\Programs\RazorCAD`, no administrator rights), for all users (Program Files), or into a folder you choose.
- Setup installs the Microsoft Edge WebView2 Runtime and .NET Framework 4.8 from Microsoft if they are missing.
- It creates Start-menu and desktop shortcuts and registers RazorCAD under Windows **Installed apps** with an uninstaller.
- The uninstaller asks before removing settings and crash reports. Part files are never touched.
- Running setup again updates an existing installation.

Silent install and uninstall:

Command	Effect
`RazorCAD-Setup.exe /S`	Silent install for the current user
`/D=folder`	Install into this folder
`/ALLUSERS`	Install for all users
`/NODESKTOP`, `/NOSTARTMENU`	Skip the desktop or Start-menu shortcut
`uninstall.exe /S`	Silent uninstall

2.3 The start window

At launch RazorCAD shows a start window with:

- **New Part** and **New Drawing**
- **Open...**
- **Sample Part**, a ready-made part to explore
- **Recent Parts**: name, folder and date. Missing files are greyed out and can be removed with ×. Parts that have an auto-save waiting are marked.
- **Recover Previous Session**, offered once when unsaved work from an earlier version is found
- A notice when new crash reports have been written

Reopen it any time with **File ▸ Recent Parts / Start...**.

2.4 The RazorCAD window

Area	What it does
Menu bar	File, Edit, View, Insert, Tools, Help. Next to the menus are quick buttons for **Save**, **Save As** and **Rebuild**.
Document tabs	Each open part or drawing has its own tab. ✕ closes a tab and asks to save.
Command ribbon	Tabs of tool buttons: **Features**, **Sketch**, **Surfaces**, **Sheet Metal**, **Fast Structure**, **Evaluate** and **Claude**. Buttons with a shortcut show the key in their bottom-right corner.
Feature tree	The part's history, with folders for Solid Bodies, Sheet Metal Bodies and Surface Bodies, the material, the planes and every feature. An eye column on the right shows or hides each item.
Configurations tab	Next to the Feature tree and the Property panel at the top of the left panel (see the Configurations chapter)
Property panel	Opens in place of the tree while you create or edit a feature. ✓ accepts (**Enter**), ✕ cancels (**Esc**).
Heads-up toolbar	Above the 3D view: view orientation, display style, selection filters, visibility (eye button) and section view
Confirm corner	The top-right corner of the view holds ✓ / ✕ for the active feature or sketch
Status line and progress bar	Bottom left. Long operations (opening, rebuilding, importing, meshing, exporting) show progress here.

2.5 Customising the ribbon

The ribbon shows its buttons in two rows.

- **Icon size**: Small, Medium or Large, from **View** ▸ **Toolbar Icons** or by right-clicking the ribbon.
- **Rearrange**: drag a button to a new place; the others slide aside.
- **Your own toolbars**: click **+** after the tabs (or right-click a tab ▸ **New Toolbar…**). Add tools by dragging a button onto its tab or with right-click ▸ **Add to "…"**. Drag a button off to remove it.
- Right-click a button on a built-in toolbar to hide it or add a separator after it. **Customize Toolbars…** shows hidden buttons again, and **Reset** puts a toolbar back as it was.

The layout is kept in `prefs.json` in RazorCAD's data folder.

3 Working with Files

3.1 Part files

Parts are saved as **.rcad** files, assemblies as **.rasm**, drawings as **.rdrw** and drawing templates as **.rdrt** files (see the Assemblies and Drawings chapters).

Command	Shortcut	Notes
File ▸ New Part	`Ctrl+N`	Opens in a new tab
File ▸ New Assembly		Opens in a new tab
File ▸ Open...	`Ctrl+O`	Opens parts, assemblies, drawings, templates, and STEP, STL, OBJ and 3MF files
File ▸ Open Sample Part		The built-in sample, in a new tab
File ▸ Save	`Ctrl+S`	Uses the normal Windows save dialog the first time
File ▸ Save As...	`Ctrl+Shift+S`	Saves a copy under a new name

3.2 Tabs and closing

Several parts and drawings can be open at once, each in its own tab. New, Open and Sample all open a new tab. Closing a tab with unsaved changes asks whether to save it, and closing RazorCAD asks to save every unsaved part.

3.3 Auto-save and recovery

Auto-save starts once a part has been saved for the first time. While the part has unsaved changes it is written next to the part file, at most every 20 seconds, as `Name.autosave.rcad`. The title bar shows the time of the last auto-save. Saving the part removes the auto-save file.

If RazorCAD closes with unsaved changes, opening the part again offers to **recover** the auto-saved version. It is also marked in the recent list.

> **Warning:** A part that was never saved is not auto-saved. RazorCAD warns before closing it. Save new parts early.

The recent-files list is kept in `%APPDATA%\RazorCAD\recent.json`.

3.4 Opening from Windows and file previews

Double-clicking a .rcad, .rasm or .rdrw file opens it in RazorCAD; if RazorCAD is already running, the file opens in a new tab there. RazorCAD sets the file types up itself when it starts. **Help ▸ Open Files With RazorCAD...** shows whether they are registered and can register them again.

Saving keeps a small picture of the view in the file. Windows Explorer, the Open and Save dialogs and the start window show it as the file's thumbnail.

A part that takes a while to build is stored, once built, in `%LOCALAPPDATA%\RazorCAD\modelcache` (kept under 1 GB). Opening the unchanged file again shows the model at once while the kernel rebuilds it in the background. Switching tabs also shows each part at once.

3.5 Importing STEP, STL, OBJ and 3MF

There are two ways to bring in a STEP, STL, OBJ or 3MF file:

- **File ▸ Open...** opens it as a part of its own in a new tab, named after the file, without asking. A STEP file that holds an assembly asks first:
  - **Open as an Assembly** makes a new assembly with the file's structure: a sub-assembly for each of its sub-assemblies, a part for each part, and a component, fixed, where the file places each one. Each part keeps its own piece of the STEP file and is built on its own, and a part used several times is built once.
  - The first build of each part is kept on disk (in `%LOCALAPPDATA%\RazorCAD\modelcache`, up to 1 GB, shared with parts). The next time the assembly opens, unchanged parts are shown straight from there instead of being rebuilt, so a big assembly reopens in seconds.
  - **Open as One Part** brings every solid into one part, as before.
- **File ▸ Import STEP / STL...** (`Ctrl+I`, also on the Features tab, and **Import STEP / STL / OBJ / 3MF** on the Evaluate tab) offers three choices: a **new part**, **insert in place** into the current part, or **insert and locate with mates**. Locating with mates adds a separate "Locate ..." Move Body feature under the import.

How imports behave:

- The file's contents are stored inside the part file, so the part does not depend on the original file.
- A STEP file's appearances come with it: the colour of each solid and of single faces, and transparency. Your own Appearance and Make Transparent / Make Opaque settings on a body win over them.
- STEP imports as an exact solid. Every solid becomes its own body; a solid made of several separate shells is split into one body per shell; loose surfaces become surface bodies.
- The bodies of a multi-body STEP file are grouped in their own folder under Solid Bodies.
- Tick **Merge touching bodies into one body** in the import feature to fuse them instead.
- STL, OBJ and 3MF meshes are converted to a solid made of their triangles, with coplanar triangles merged. A 3MF file's units and build placement are read. A mesh that is not closed comes in as a surface body, with a note saying how many open edges it has; **Repair Mesh** or **Convert to Solid** (Surfaces chapter) can close it.
- Big meshes are simplified on import, by default to 80,000 triangles, keeping their edges and outlines. Set the limit in the import's properties (**Simplify meshes above ... triangles**; 0 never simplifies). The import's note says how far a mesh was simplified.
- Imported bodies can be cut, extended, patterned and (for STEP) filleted like any other body. A cut or fillet on a body of more than 20,000 mesh faces warns that it can take several minutes.

IGES files can't be read: the geometry kernel is built without IGES. Use STEP, which every CAD program writes.

3.6 Exporting

Command	Result
File ▸ Export ▸ STEP...	An exact B-rep. Each body is written as its own named, coloured solid.
File ▸ Export ▸ STL...	A binary mesh
Evaluate ▸ **Export OBJ** / **Export 3MF**	A mesh; 3MF is written in millimetres
Right-click a body ▸ **Export Body as STEP...**	One body only

Command	Result
File ▸ Export ▸ Face as DXF…	The outline of a flat face (see below)
File ▸ Export ▸ Flat Pattern as DXF…, Sheet Metal ▸ **Export DXF**	Flat patterns (see the Sheet Metal chapter)
Drawing ▸ **Export PDF** / **Export DXF**	Drawing sheets (see the Drawings chapter)

3.6.1 Export Face as DXF

Click a flat face, then right-click ▸ **Export Face as DXF…** (or **File ▸ Export ▸ Face as DXF…**). The face's outline (layer `OUTLINE`) and holes (layer `HOLES`) are written as a 2D drawing in millimetres, seen from outside the part. Use it for laser or waterjet cutting, or to take a profile into a 2D CAD program.

Lines, arcs and circles are exact; other curves become fine polylines. The file is DXF R12, which practically every program can read.

4 Navigating the 3D View

4.1 Mouse and keyboard

Action	Mouse / keys
Rotate (free, about the model centre)	Middle-drag, or left-drag on the model outside a sketch
Pan	**Shift** + right-drag, or **Ctrl** + middle-drag
Zoom	Mouse wheel (toward the cursor), or **Shift** + middle-drag
Roll	**Alt** + middle-drag, or **Alt** + ← / →
Rotate in steps	Arrow keys 15°, **Shift** + arrows 90°
Pan in steps	**Ctrl** + arrow keys
Zoom to fit	**F**
Standard views	**Ctrl+1** … **Ctrl+7**
Normal To	**Ctrl+8** (press again to flip to the back side)

4.2 Standard views

View menu: Front (**Ctrl+1**), Back (**Ctrl+2**), Left (**Ctrl+3**), Right (**Ctrl+4**), Top (**Ctrl+5**), Bottom (**Ctrl+6**), Isometric (**Ctrl+7**) and Trimetric. **Normal To** (**Ctrl+8**) looks straight at the selected face or plane.

4.3 Display styles

View menu or the heads-up toolbar: **Shaded With Edges**, **Shaded**, **Hidden Lines Removed** and **Wireframe**.

4.4 Visibility

- **View** ▸ **Show / Hide Planes** toggles the standard planes.
- The **eye column** at the right of the Feature tree shows or hides each sketch, plane, axis, point, curve, body and body-making feature on its own. Hiding a feature hides the bodies it built. Planes, axes, points and curves can also be hidden one at a time with right-click ▸ **Hide**.
- The **eye button** on the heads-up toolbar (or **View** ▸ **Visibility…**) filters what is shown: sketches, axes, planes, points and surfaces, with select-all and deselect-all.
- Right-click a body (tree or view) ▸ **Hide**. **Tab** hides the body under the cursor; **Shift+Tab** shows the hidden body under the cursor.
- **View** ▸ **Show Hidden Bodies** shows hidden bodies so you can click them to show them again. Choose **Done Showing Hidden Bodies** to finish.
- Right-click a body ▸ **Isolate** hides everything else. The isolated body can still be edited. **View** ▸ **Exit Isolate** returns.
- Right-click a body, or any feature that made or changed it, ▸ **Make Transparent / Make Opaque**. Transparency is saved with the part and can be undone.

4.5 Section view

Tools ▸ Section View... (or the Evaluate tab) cuts the model open on a plane so you can see inside. Choose the plane, an offset (equations are allowed) and flip the side kept. Each cut body or component gets its own cap in its colour, outlined and hatched with section lines; the hatch angle and spacing change from body to body so touching parts read apart. Close the Property panel to remove the section. Section View can't be used inside a sketch.

5 Selecting

5.1 Selecting in the part

A click selects what is under the cursor, in this order: a **vertex**, then an **edge**, then a **face**, then a **plane**. **Ctrl**+click adds to the selection.

5.2 Selection filters

The filter buttons on the heads-up toolbar limit what can be picked. Several filters can be on at once, and they also apply while picking references for features.

Filter	Key	Picks
Faces	**X**	Model faces
Edges	**E**	Model edges
Vertices	**V**	Corners (edge end points)
Solid Bodies	**B**	Whole bodies (a click on any face picks its body)
Planes		Standard and reference planes
Clear filters	**F5**	Back to picking anything

5.3 Quick readout of sizes

Clicking an edge shows a small tag in the view: its length, R and length for an arc, or Ø and circumference for a circle. **Ctrl**+click more edges for their total length (the perimeter when they go all the way round). Sketch entities shown in the view can be clicked the same way.

One face shows its area (and Ø when round). **Ctrl**+click a second face for how they lie: "Parallel · 12 mm apart", "Perpendicular · 90°" or the angle; two round faces give their centre distance; three or more faces their total area.

5.4 Highlighting from the tree

Hovering a feature in the Feature tree highlights what it made in orange: the faces of a boss, cut, fillet, chamfer, shell, pattern or moved body; a sketch's curves; a plane, axis or point; a surface feature's surfaces. Clicking it keeps the highlight in blue while it is selected.

5.5 Selecting hidden bodies

Hold **Shift** and **right-click** to step through the bodies and surfaces under the cursor, front to back; **Shift** + left-click steps back. The current one is shown in orange, through whatever is in front of it. Release **Shift** and **left-click** on the same spot to select it. **Esc** or a click elsewhere cancels.

- With only the face filter on, or while a Property panel asks for faces (Shell, Offset Surface), it steps through faces instead.
- While a Property panel asks for bodies (Combine, Split, Move/Copy, Knit, Thicken), the click adds the body.

5.6 Right-click menus

Right-click on	Offers
A face	Sketch, Normal To, Export Face as DXF, Fillet, Chamfer, Appearance…
An edge	Fillet, Chamfer (started with that edge), Reference Axis
A vertex	Reference Point
A face, edge or body	Body Properties, Body Appearance…, Export Body as STEP, Hide, Isolate, Make Transparent, folder commands
A sketch in the tree	Edit, Show / Hide, Change Sketch Plane…
A sheet metal body	View Flat Pattern, Export Flat Pattern as DXF
Empty space	Show Hidden Bodies, Measure and view commands

6 Sketching

6.1 Starting and ending a sketch

Click a plane or a flat face, then click **Sketch** (Sketch tab), or right-click a face ▸ **Sketch**. Clicking a sketch tool such as **Circle** with no sketch open also starts a 2D sketch: pick a plane or flat face and the tool is ready. (For sketches in space, see the 3D Sketches chapter.) You can also start a feature such as Extrude without a sketch: RazorCAD asks you to pick a plane or face, you sketch, and when you exit the sketch the feature continues. If the profile is open you can keep editing or abort and keep the sketch.

- **`Esc`** ends the active sketch tool first; pressing it again closes the sketch and keeps it.
- **`Enter`** does **not** leave the sketch.
- The ✓ in the confirm corner, or **Exit Sketch**, closes the sketch.
- **`Ctrl+Z`** / **`Ctrl+Y`** undo and redo inside the sketch.

6.2 The sketch tool wheel

While sketching, press and hold the right mouse button. After a moment, or as soon as you start to drag, a wheel of sketch tools appears around the pointer. Keep the button held, move toward a tool (it highlights) and release to activate it. Releasing in the centre or pressing **`Esc`** cancels. A quick right click (press and release without moving) still opens the right-click menu.

6.3 Defined and under-defined sketches

Sketches are made of points, lines, arcs and circles held in place by **relations** and **dimensions**. A numerical constraint solver keeps them satisfied.

- Geometry is **blue when under-defined** and **black when fully defined**.
- The Property panel shows the remaining degrees of freedom.
- An under-defined sketch shows **(-)** in the tree. Construction lines only need to be positioned.

6.4 Sketch orientation

In the sketch Property panel, **Rotate 90°** turns the sketch axes about the sketch normal, and **Flip Side** looks at the sketch from the other side of the plane. Both are saved with the sketch.

6.5 Sketch tools

Tool	Key	How it works
Select	**S**	Select entities, drag geometry
Line	**L**	Click a chain of points. Coincident, horizontal and vertical relations are added automatically; snapping to a midpoint or onto a curve adds those relations too. Double-click, **`Enter`** or **`Esc`** ends the chain.
Centerline		Construction line for symmetry, mirroring, gears or as a revolve axis
Corner Rectangle	**R**	Built with horizontal and vertical relations
Center Rectangle		Also adds construction diagonals with a midpoint relation
Circle	**C**	Centre, then radius

Tool	Key	How it works
Centerpoint Arc	`A`	Centre, start, end; the direction follows the mouse
3 Point Arc		Start, end, then a point it runs through
Tangent Arc		From the end of a line or arc, kept tangent by a Tangent relation
Ellipse		Centre, end of the major axis, then the minor radius. Trim it for an elliptical arc.
Spline	`N`	See below
Point		Lone sketch points that stay, for example to place holes
Straight Slot		Built with tangent and equal relations
Polygon	`P`	Set the number of sides in the Property panel
Dimension	`D`	See below
Trim	`T`	See below
Extend		A line or arc end runs on to the first entity in its way and stays on it
Split		Splits a line or arc in two at the click; the halves stay collinear or equal
Project Edges	`E`	See below
Offset	`O`	See below
Mirror		Select the entities plus a centerline, then click Mirror
Import DXF		See below
Linear / Circular Sketch Pattern		Copies the selected entities with the relations and dimensions among them. A selected point is the circular pattern's centre.
Make Block / Insert Block		Make Block saves the selection (with its relations) under a name in your settings. Insert Block places copies with a click, in any sketch, and can delete blocks from the library.
Text		See below
Gears		See the Gears chapter
Airfoil		See the Airfoils chapter
Sketch Fillet / Sketch Chamfer		Set the radius or distance, then click corners. The first one made gets the dimension; the others made before you leave the tool are made Equal to it, so one value changes them all.
Intersection Curve		Projects the outline where the sketch plane cuts the bodies
Construction Geometry	`Q`	Toggles the selected curves to or from construction (drawn dotted, ignored by features)
Display Relations		Shows or hides the relation glyphs
Delete	`Del`	Deletes the selected entities or relations

6.6 Dimension

Click a line, circle or arc (or two entities), then click to place the dimension.

- Moving the mouse switches between horizontal, vertical and aligned.
- Two lines give an angle; a point and a line give a perpendicular distance.
- Values accept equations, such as `width/2`.

- A dimension that would over-define the sketch is added as a driven (reference) dimension.
- **Circles and arcs:** dimension from a circle or arc to another circle, arc, point or line by clicking both. The Modify box offers **Center** (centre to centre), **Min** (nearest) and **Max** (farthest). Min also measures the gap between concentric circles. Switching keeps the geometry in place and shows the new value.
- **Outside references:** you can also click model edges, model vertices or entities of earlier sketches. RazorCAD creates linked construction geometry and dimensions to it, so the dimension follows the model. Example: "keep this hole 25 mm from the right edge" still holds after the width changes.

6.7 Trim

Power trim: hover over a segment and it turns red; click to remove it up to the nearest intersections. Lines are shortened or split, circles become arcs, and new end points get on-curve relations.

6.8 Offset

Click a line, arc or circle and its whole connected chain is offset. Move the mouse to the side and distance you want and click, then type the exact distance. Corners are re-trimmed or extended.

- You can also click a **model edge** (with **Select chain**, the connected edges lying in a plane parallel to the sketch, such as a face outline) or a curve of another sketch.
- **Linked offset** (default): the offset follows its source, including model edges when the model changes, and one dimension drives the chain (relation "Offset ⇉").
- Untick it for a **single offset**: plain geometry with no link.
- While the tool is active, the faces the sketch plane cuts are highlighted and the cut outline is drawn in teal. Clicking it offsets the whole outline.

6.9 Project Edges

Click a model edge, or an entity of an earlier sketch, to project it into this sketch. Or select edges first and then click **Project Edges** (or press **E**): they are projected at once. Projected geometry is drawn in **purple**, is linked and updates when the source changes (relation "On Edge ⇲"). The ends of projected lines are joined to each other and to sketch geometry already there, so the edges of a face give a closed outline ready to extrude. While the tool is active, the outline where the sketch plane cuts the bodies is drawn in teal; clicking it converts the whole outline. This works for solids and surfaces.

6.10 Spline

Click the points the curve runs through (they snap like line points), then double-click, **`Enter`** or **`Esc`** to finish, or click the first point again to close it. Drag a point to reshape the curve, or drag the curve to move it; deleting a point takes it out of the spline. Splines make profiles on their own or with lines and arcs and are exact curves for extrudes, revolves, cuts, lofts and sweep paths. Lines can be trimmed where they cross a spline.

6.11 Import DXF

Import DXF brings a DXF file into the open sketch: lines, arcs, circles, polylines with arcs, ellipses and elliptical arcs, splines, points and blocks. The file's units are converted to millimetres. You can scale it and place it as in the file, centred, or by its corner.

6.12 Text

Click **Text** on the Sketch tab, then click to place. The panel sets the text, font (the PC's installed fonts), height (capital-letter height), angle, bold and italic. Drag the text to move it. Text extrudes as a boss or cut; text inside an outline is cut out of it. To put text on a curved face, use **Wrap** (Features chapter).

6.13 Relations

Select entities (**Ctrl**+click for several). The Property panel lists the relations that apply:

Selection	Relations offered
1 line	Horizontal, Vertical, Fixed
2 or more points	Coincident, Horizontal, Vertical
2 or more lines	Parallel, Perpendicular, Collinear, Equal, Horizontal, Vertical
Point + line	Coincident (on line), Midpoint
Point + circle/arc	Coincident (on curve)
Line + circle/arc	Tangent
Circles/arcs	Equal, Concentric, Tangent
2 points + line	Symmetric (about the line)
3 lines	Symmetric (two lines about a centerline)

Relations are shown as small green glyphs next to the geometry. Click a glyph to select it, then press **Del** to remove it. Hovering a glyph highlights the entities it ties together. Redundant or conflicting relations are refused with a message.

6.14 How profiles are found

Profiles are found automatically from all non-construction geometry and sent to the kernel as exact lines, arcs and circles. Intersections are split, dangling lines are ignored and nested loops become holes. A full circle (or arcs that make one) becomes a single closed curve, so an extruded circle has one continuous cylindrical face.

6.15 Showing and hiding sketches

Sketches absorbed by a feature are hidden by default. Right-click a sketch in the tree ▸ **Show / Hide**, or use its eye in the tree.

6.16 Change Sketch Plane

Right-click a sketch ▸ **Change Sketch Plane…** and pick another plane or flat face for it, for example to repair a sketch whose plane was deleted ("The reference plane is missing"). Its curves keep their sketch coordinates. Only planes above the sketch in the tree are offered.

7 3D Sketches

A **3D sketch** holds lines, arcs, circles and splines in space rather than on one plane. Use it for frames (see the Fast Structure chapter), pipe, duct and wiring routes, and sweep paths.

7.1 Starting and ending a 3D sketch

Click **3D Sketch** on the Sketch tab or the Fast Structure tab (or **Insert ▸ 3D Sketch**). The sketch opens empty with the Line tool ready; there is no plane to pick. While it is open the **3D Sketch** button stays highlighted and the first button becomes **Exit 3D Sketch**. Tools that only work in 2D sketches are greyed out. **Ctrl+Z** / **Ctrl+Y** undo and redo inside the sketch.

7.2 Drawing in space

- Click points to draw lines. A line snaps along **X, Y or Z** from the last point whenever the mouse moves along that direction on screen, in any view. A dashed line shows it and an Along X/Y/Z relation is added. Hold **Ctrl** or **Shift** to draw freely.
- Away from the axes, lines follow the **drawing plane** through the last point. **Tab** cycles XY → YZ → ZX; the plane's name shows at the cursor.
- **Ctrl**+click a plane or flat face, or click a plane in the tree, to draw on it. Points placed there get an **On Plane** relation. **Tab** or **Esc** goes back to XY / YZ / ZX.
- Points snap, in this order, to the sketch's own points, to line midpoints (Midpoint relation), to a point along a line (On Line relation), to the origin and model vertices, along X/Y/Z, then to the drawing plane.
- The **Next Point** box in the Property panel takes exact offsets, for example `0, 0, 250`.
- A selected point shows its X, Y and Z; type new values to move it. A selected line shows its length, ΔX/ΔY/ΔZ and end points; an arc its radius, angle and length.

7.3 3D sketch tools

Tool	Key	How it works
Line	**L**	As above
Centerline		Construction lines that shape the sketch (for example a frame envelope) but make no curve
Point		A lone point
Circle, **Centerpoint Arc**, **3 Point Arc**, **Polygon**	**C**, **A**, **P**	Drawn on the active drawing plane through the first click. **Tab** changes the plane before or between clicks. A polygon stays regular.
Tangent Arc		Click the free end of a line, arc or spline, then where the arc ends. Keep clicking for more arcs; **Esc** ends.
Spline	**N**	Click the points it runs through; double-click or **Enter** ends it, clicking the first point closes it. A Tangent relation with a line at an end makes it leave the line smoothly.
Sketch Fillet / Sketch Chamfer		Click a corner where two lines meet. The sharp corner stays as a point on both lines, so lengths measured to it keep their meaning.

Tool	Key	How it works
Trim	`T`	Click the piece to remove, up to where other entities cross or touch it
Project Edges	`E`	Click model edges to bring them in, linked to the model (see below)
Dimension	`D`	Length, distance between points, angle between lines, arc radius, circle diameter. While placing a length, **Tab** cycles true length → ΔX → ΔY → ΔZ.
Construction Geometry	`Q`	Toggles the selected lines to or from construction
Delete	`Del`	Deletes the selection

Move / Copy / Rotate: drag a selected line, arc or circle to move the selection (it snaps along X, Y and Z), or type ΔX/ΔY/ΔZ and click **Move** or **Copy**, or rotate about X, Y, Z or the drawing plane's normal with **Rotate** or **Copy Rotated**. Copies keep the relations among the copied geometry. Fixed and projected geometry stays put.

7.4 Relations in 3D

Along X/Y/Z, Parallel, Perpendicular, Collinear, Equal Length, Equal Radius, Tangent, Midpoint, On Line, Parallel to XY / YZ / ZX, On XY / YZ / ZX plane, On Plane, Regular Polygon, Fix and Merge (coincident). A solver keeps everything held and reports **Fully defined** or the degrees of freedom left. A relation or dimension that conflicts is refused with a message.

7.5 Projected edges in 3D

Project Edges brings model edges into the 3D sketch: lines stay lines, circular edges become arcs or circles and other curves become splines, with shared corner points. The geometry (purple) stays linked to the model: when the part changes, it follows the edges, and lines and dimensions attached to it follow too. **Unlink from the Model** in the Property panel makes it ordinary geometry.

7.6 Curves from a 3D sketch

Each connected run of lines, arcs and splines is one curve; a T-junction splits it into separate runs. A run can be used as straight segments with an optional bend radius at the corners (pipe and duct routes) or as a smooth spline, and can be closed into a loop. **Sweep**, **Swept Surface**, **Lofted Surface**, **Curve Driven Pattern** and **Structural Member** use 3D sketches.

8 Gears

RazorCAD's **Gears** tool on the Sketch tab draws involute gear trains as closed profiles, ready to extrude.

8.1 Creating a gear train

1. Draw lines between the gear centres (centerlines work well).
2. Select the lines (or a single point for one gear) and click **Gears**. If you are not in a sketch, RazorCAD opens the selected sketch (or starts one) and asks you to click the centre lines, then press **`Enter`** or **Make Gears**.
3. Each line end point becomes a gear. Neighbouring gears mesh along the line, and teeth are turned automatically so they mesh.
4. Extrude the sketch: each gear becomes its own body.

Gear numbers are shown on the gears in the sketch (view labels only, not geometry).

8.2 The Gear Train panel

Click a gear to open it.

Setting	Meaning
Module and pressure angle	The tooth size and shape for the whole train
Teeth per gear, or ratio per mesh	Setting a ratio sets the next gear's teeth. Changing teeth or a ratio changes the centre-line lengths to suit, including your own length dimensions on those lines.
Backlash, tip/root clearance	Tolerances for prototyping and 3D printing
Bore diameter	The centre hole
Turn	Rotates the first gear
Set the line lengths to the mesh distance	Adds driving dimensions so the gears mesh exactly; each mesh shows ✓ or how far off it is

8.3 Rules for a valid train

Centres must be exactly at the mesh distance, gears must not overlap and each gear needs at least 6 teeth. Fixed line lengths get a module and teeth that fit. Edits that would break a train are refused. A broken train is drawn red with a **Fit gears to the lines** button and can't be extruded.

9 Airfoils

The **Airfoil** tool on the Sketch tab draws a wing or blade section on a line.

9.1 Drawing an airfoil

1. Select a line in a sketch and click **Airfoil** (or click **Airfoil**, then the line).
2. The line becomes the **chord**: a construction line with the leading edge at its start. The airfoil is drawn on it.
3. The airfoil follows the line. Drag its ends or dimension it (or set **Chord length** in the panel) and the airfoil scales and turns with it.
4. Extrude the sketch. The profile is one smooth closed curve that follows the exact section to within 0.002 % of the chord.

9.2 Sections

Choice	Sections
Library	28 NACA 4-digit sections (0006 to 6412), 10 NACA 5-digit sections (21012 to 25012, and the reflexed 23112), and Clark Y (11.7 %, 8, 10 and 14 %)
NACA code	Type any code, for example `4415` or `23012`
Custom	4-digit: maximum camber, its position and thickness. 5-digit: design lift coefficient, camber position, reflexed camber line and thickness. Clark Y: thickness. Decimals are allowed.

Options: sharp or blunt trailing edge, camber to the other side, and the leading edge at the other end of the line.

10 Solid Features

10.1 Live previews

While a feature's Property panel is open, the model is rolled back to just before that feature and the feature is shown in colour: **yellow** for material it adds and **red** (seen through the body) for material it removes. Fillets and chamfers show the material they add or remove; reference geometry being edited is drawn in orange. Click ✓ (or **Enter**) to finish.

Every value field accepts numbers or equations (see Equations in the Evaluate chapter).

10.2 Extrude and Extrude Cut

Extrudes a sketch profile into a solid, or cuts it out.

End conditions:

End condition	Meaning
Blind	A set depth
Mid Plane	The depth split equally on both sides of the sketch
Through All / Through All – Both	Through every body, in one or both directions
Up To Vertex	Click a model vertex or choose a reference point
Up To Surface	Click a flat model face or choose a plane; angled surfaces are supported and the extrusion is trimmed to the surface
Offset From Surface	As Up To Surface, at a set distance from it

Options:

- **Merge result** (boss): untick to keep the new material as a separate body even where it touches. A boss that doesn't touch existing material makes a new body anyway.
- **Thin Feature**: makes a wall of the given thickness from the profile; direction one side, other side or mid-plane. Open sketches switch it on automatically.
- **Feature Scope** (cuts): **All bodies** by default; untick it and click the bodies to cut.

10.3 Revolve and Revolve Cut

Revolves a profile about an axis: the sketch axes, any line in the sketch (such as a centerline), a reference axis, a straight model edge or a cylindrical face (click it in the view). Cuts have the same Feature Scope option as extruded cuts.

10.4 Sweep and Sweep Cut

Sweeps a profile along a path.

- **Profile**: a sketch, or **Circular** with a diameter (useful for springs).
- **Path**: a sketch chain or a helix.
- **Thin** option.

10.5 Loft and Loft Cut

Blends between two or more profile sketches, in order. A third or later profile shapes the middle.

- A circle is split into as many equal arcs as the other profile has corners.
- Profiles with different segment counts give an error naming the sketches.
- Options: straight (ruled) sides, Thin.

10.6 Hole

Hole (Features tab, **Insert** ▸ **Features** ▸ **Hole...**) makes standard holes.

Type	Sizes
Hole (clearance)	ISO 273 close, normal or loose fit, for M2 to M24
Counterbore	For socket head cap screws (ISO 4762)
Countersink	ISO 10642, 90°
Tapped hole	ISO metric or Unified (inch), drilled at the tap drill size; the thread is kept for drawing callouts
Custom sizes	Your own diameters and depths

- **Blind** with a 118° drill point, or **Through All**; reversible.
- Place holes at clicked points on a flat face, or exactly at the points and circle centres of a sketch on the face.
- Holes work on sheet metal too, and a drawing's **Hole Callout** reads their size from the feature.

10.7 Helix

Pick a sketch with a circle; it sets the axis and diameter. Define the helix by **pitch and revolutions**, **pitch and height**, or **height and revolutions**. Options: start angle, clockwise, reverse. Use a helix as a sweep path for springs and coils.

10.8 Thread Cut and Thread Extrude

Click a bore (Thread Cut) or a shaft (Thread Extrude).

- The size is chosen automatically from the diameter, or choose a **metric** (ISO) or **inch** (Unified) standard size, or custom. Coarse / UNC sizes are preferred.
- Optional pitch; **left-hand** option.
- Length: the whole face, a depth, or up to a plane or face. Threads never reach past the ends of their face.
- The bore or shaft is resized to the thread size if needed (a bore to the minor diameter, a shaft to the major). A bore that would break out of the body is refused.
- Threads run all the way to the ends: grooves run out into open air, otherwise the thread is cut off flush with the end.

10.9 Wrap (Emboss / Deboss)

Pick a sketch (outlines and/or text) and a face, flat or curved. The sketch is projected along its normal onto the face and raised (emboss) or sunk (deboss) by the depth.

10.10 Patterns and Mirror

Patterns and mirrors copy features or bodies. Any feature that changes the bodies at a place can be copied: extrudes, cuts, holes, fillets, chamfers, drafts, domes and sheet metal forms. A feature without a shape of its own (such as a fillet) is run again on the matching edges or faces of each copy.

Feature	Copies
Linear Pattern	Along one or two directions (X, Y, Z or any reference axis)
Circular Pattern	Round an axis (X, Y, Z or any reference axis)
Mirror	About a plane or flat face
Sketch Driven Pattern	At every point (or circle centre) of a sketch
Curve Driven Pattern	Along a sketch, 3D sketch or helix, spaced evenly to the end or at a set distance; moved, or turned to follow the curve
Fill Pattern	On a square or staggered grid over a flat face, kept a margin from its edges (for example a perforated panel)

Patterns of cuts are cut out in one operation, so large patterns rebuild quickly.

A pattern or mirror of a sheet metal feature that makes a body (Base-Flange, SM-Extrude, Lofted-Bend, Transition) copies that body as it is at the pattern, with the flanges and cuts added to it so far. Each copy is sheet metal with its own flat pattern. A mirrored copy's flat pattern is the mirror image, with its bends going the other way. A mirrored lofted bend becomes a plain solid.

10.11 Fillet

Open the feature and click edges on the model. The model is rolled back to just before the feature so every edge can be picked; tick **Preview** to see the result. The radius accepts equations.

- Edges are remembered by type, direction and position, and found again after parameter changes.
- Where a fillet can't be added on top of earlier fillets (a corner where several blends meet), RazorCAD recomputes it together with the preceding fillets in one operation.
- Where two blends exactly meet (for example R2 on both edges of a 4 mm wall) the result is a full round.
- **Variable size**: a radius at the start and another at the end of each edge.
- **Face fillet**: pick faces instead of edges; the edges where they meet are rounded.

10.12 Chamfer

Click edges as for a fillet. Chamfer types: **equal distance**, **distance + angle** or **two distances**. **Flip direction** swaps the sides.

10.13 Shell

Click the faces to remove and set the wall thickness, inward or outward. Where a fillet or chamfer is smaller than the wall, the outside keeps it and the inside gets a sharp corner.

10.14 Draft

Pick a neutral face, then the faces to tilt by an angle. **Reverse** tilts them the other way.

10.15 Rib

An open sketch line is thickened and filled out until it meets the model, either in the sketch plane (for example a gusset across a bracket corner) or square to it.

10.16 Dome

A spherical cap or half ellipsoid on round faces, or a smooth dome filled over any other flat outline. **Reverse** presses it in.

10.17 Move Face

Push or pull flat faces along their normal, or turn a face about one of its edges. The neighbouring faces follow.

10.18 Scale

Scales every body by one factor, about the origin or about the centre of the bodies.

10.19 Long operations

If a feature takes more than a few seconds, a banner above the model says what is being worked on ("Still working on Extrude Cut1 (12 s)…") with a **Stop** button (also next to the progress bar). Stop restarts the geometry kernel: a feature you are still editing is cancelled and a finished feature is suppressed, so you can unsuppress it to try again.

11 Bodies and Multi-Body Parts

11.1 Solid bodies

A part can contain several solid bodies. A boss that doesn't touch the existing material makes a new body; untick **Merge result** to keep a touching boss separate. A cut that slices a body in two leaves two bodies.

The **Solid Bodies (n)** folder at the top of the tree lists the bodies:

- Click a body to select and highlight it.
- Double-click (or right-click ▸ **Rename**) to name it. Unnamed bodies are named after the last feature that changed them.
- Bodies are identified by the feature that created them, so names and materials stay attached when you change dimensions.
- **Folders**: right-click ▸ **Add to New Folder** or **Move to folder** to organise bodies.
- **Identical bodies** are listed once with their count (×N). Open the item to see each body. Right-click for Isolate, Hide or Show all of them. Identical means the same volume, surface area, principal moments of inertia, handedness and material (and for sheet metal, the same thickness and flat pattern).

11.2 Body Properties

Right-click a body (tree or view) ▸ **Body Properties...**. It holds the body's name, **material**, description and any number of **custom properties** (name/value pairs such as part number or finish). It also shows the body's mass, volume, area and centre of mass. For sheet metal bodies it adds the sheet data (Sheet Metal chapter).

11.3 Materials

The **Material** item in the tree sets the part material (**Tools** ▸ **Edit Material...**). A body set to **Use part material** follows it. A body given its own material keeps it when the part material changes and is drawn in that material's colour. Mass Properties adds up every body with its own density. The material list is in Appendix B.

11.4 Appearance

Right-click a face ▸ **Appearance...** or a body ▸ **Body Appearance...** to give it a colour, or **Default colour** to set it back. Face colours stay with their face when the model changes. A body's own appearance wins over its material colour and over colours imported from a STEP file.

11.5 Exploded views of a part

A part with two or more bodies can have exploded views: **Insert** ▸ **Exploded View...**, or right-click the Solid Bodies folder. Only the display moves; the model stays as it is. The steps are made as for an assembly (see Exploded views in the Assemblies chapter).

11.6 Combine and Subtract Bodies

Combine joins bodies with **Add**, **Subtract** or **Common**; the first body clicked is the main body. **Subtract Bodies** picks a cutting body and target body; **Consume cutting body** (on by default) removes the cutting

body afterwards.

11.7 Split

Cuts bodies into separate bodies with a plane, a flat face or a surface body. With a plane or face you can keep all the pieces or just the piece on one side.

11.8 Move/Copy Bodies

Moves or copies solid and surface bodies.

- **By coordinates**: ΔX/ΔY/ΔZ plus a rotation about X/Y/Z or a reference axis, through the axis or the bodies' centre. Copies repeat the step.
- **With mates**: Coincident, Distance, Parallel, Perpendicular, Angle and Concentric between an entity of the moving bodies (face, edge, vertex) and fixed geometry (another body, a plane, axis or point). **Flip alignment** turns a mate around.

11.9 Delete Body

Removes a body from the model from that point in the history. Deleting the Delete Body feature brings the body back.

12 Surfaces

The **Surfaces** tab builds surface bodies (zero-thickness faces), listed under **Surface Bodies** in the tree. Cuts and Combine only act on solid bodies; Knit, Thicken or Convert to Solid turn surfaces into solids.

12.1 Making surfaces

Feature	What it does
Extruded Surface	Sweeps the sketch's curves in a straight line; open chains are fine
Revolved Surface	Revolves the sketch's curves about an axis
Planar Surface	Fills a closed sketch (holes allowed)
Lofted Surface	A surface through two or more profiles (sketches or 3D sketches), open or closed, optionally ruled (straight between profiles). Open profiles drawn in opposite directions are lined up automatically.
Swept Surface	An open or closed sketch profile, or a circle, swept along a sketch, 3D sketch or helix
Filled Surface	A smooth patch over a hole bounded by edges of a surface or a solid, touching or optionally tangent to the faces round it. It can be knit into the surface and close a solid. A flat loop gives a flat patch.
Boundary Surface	A surface through curves in two directions. The first and last curve of each direction bound it and must meet at their ends; the curves between shape it.
Ruled Surface	Straight-sided surfaces standing on edges, either normal to the surface or along a direction tapered by an angle, on one or both sides (for example mould parting surfaces)
Offset Surface	Offsets faces of any body
Copy Surface	Copies faces of any body as surface bodies, leaving the body unchanged. **Join copies that touch into one surface body** is on by default.

12.2 Changing surfaces

Feature	What it does
Trim Surface	Cuts surfaces with a plane, a surface body or a sketch's curves (square to the sketch). Click the pieces to keep, or to remove. **Mutual** trims the tool surface too.
Extend Surface	Carries open edges straight on by a distance, following the surface's slope at the edge, and knits the extension on
Untrim Surface	Fills a surface's holes, or gives it back its natural extent (holes and outline). Pick faces to untrim only those.
Knit Surface	Sews surfaces together within a gap tolerance. When they enclose a volume, **Try to form a solid** turns them into a solid.
Thicken	Gives a surface a wall thickness (one side or both) and makes it a solid
Delete Face	Takes faces out of a solid or surface. Either leave the hole (the body becomes a surface body) or patch each hole smoothly, optionally tangent, so the solid stays solid. Good for removing fillets and bad faces from imported parts.
Replace Face	Moves flat faces of a solid onto a surface body, so the body grows or shrinks to meet it (for example a curved top). The surface must reach past the faces all round.
Flatten Surface	Unrolls faces of a surface or solid flat beside the model as a surface body (see below)

Feature	What it does
Split (with a surface)	Splits solid bodies with a surface body
Move/Copy Surfaces	As Move/Copy Bodies, for surfaces
Delete Surface	Removes a surface body

12.3 Curves and split lines

Feature	What it does
Split Line	Splits faces along a sketch's projection, or where a plane or surface crosses them. The body stays one body with its faces split, for example for a parting line, a draft or a colour.
Projected Curve	Projects a sketch along its normal onto faces, as a 3D curve
Composite Curve	Joins model edges end to end into one curve

Projected and composite curves can be used as sweep paths, lofted surface profiles, curve-driven pattern paths and boundary surface curves.

12.4 Flatten Surface

Click the faces to unroll. Faces that unroll (cylinders, cones, bent sheets) come out exact. A doubly curved face (for example a dome) can't lie flat without stretching: it is spread out keeping its edge lengths as well as possible, and the note gives the size, the area change and how much the edge lengths changed. **Relaxing passes** sets how hard it tries. A closed tube needs a seam (Split Line) to open out.

12.5 Meshes and closing surfaces

- **Convert to Solid** (also right-click a surface body): surface bodies that together enclose a volume, or a mesh, are joined (edges closer than the gap tolerance meet) and made into one solid. When they leave gaps it says how many open edges and holes there are.
- **Repair Mesh**: closes the holes of a surface body or a mesh with missing pieces (flat holes with a flat face, curved holes with a filling surface, holes in a mesh with triangles), turns triangles that face the wrong way round, and makes a solid. Option: the largest hole to close.

Surface quality can be checked with Zebra Stripes, Curvature and Draft Analysis (Evaluate and Tools chapter).

13 Reference Geometry

13.1 Reference Plane

Pick up to three references (faces, edges or vertices on the model; planes in the view; planes, axes or points in the tree). The plane type follows from what you pick. Click a reference again to remove it.

References	Plane made
One plane or face	Offset (distance, flip)
Two planes or faces	Mid plane: halfway between parallel faces, or the bisector of angled ones
Plane/face + edge/axis	Through the edge at an angle (Parallel 0° and Perpendicular 90° buttons)
Edge/axis (+ optional point)	Normal to the edge, at its start, middle, end or the point
Plane/face + point/vertex	Parallel, through the point
Three points or vertices	Through three points

13.2 Reference Axis

Made from two planes, two reference points, a straight or circular model edge, a sketch line, or a cylindrical face (axis at its centre). Right-click an edge ▸ **Reference Axis** to start one. Axes are drawn dashed and follow the model when it changes. A reference axis can drive a revolve or a pattern.

13.3 Reference Point

Coordinates (equations allowed), a model vertex, an arc centre or an edge midpoint. Right-click a vertex ▸ **Reference Point** to start one.

14 The Feature Tree and History

14.1 Editing features

- Double-click a feature, or right-click ▸ **Edit Feature**, to open its Property panel.
- Select a feature and press **`Del`** (or **Edit** ▸ **Delete**) to delete it.
- **Edit** ▸ **Undo / Redo** (**`Ctrl+Z`** / **`Ctrl+Y`**) works on the whole part history.
- **Rebuild** (**`Ctrl+B`**, or the quick button) rebuilds the part.
- A feature that fails shows an error in the tree so you can fix or re-pick its references.

14.2 Rollback bar

The blue line at the bottom of the tree. Drag it up, or right-click a feature ▸ **Roll Back** / **Roll to End**. Features below the bar are suppressed, and new features are inserted above it.

14.3 Reordering features

Drag a feature up or down the tree. Its absorbed sketch moves with it. A move that would break a reference is refused with the reason.

14.4 Reference arrows

Select a feature in the tree. Blue arrows come from what it uses; purple arrows go to what uses it.

14.5 Tree symbols

Sheet metal features and bodies have their own icons. Under-defined sketches show **(-)**. Sketches absorbed by features are hidden by default.

15 Sheet Metal

The **Sheet Metal** tab holds the sheet metal features. Sheet metal bodies are listed in the **Sheet Metal Bodies** folder below Solid Bodies, with sheet metal icons.

Group	Features
Making a sheet	Base Flange, Sheet Metal Extrude, Lofted Bend, Transition / Reducer, Convert to Sheet Metal
Flanges and bends	Edge Flange, Miter Flange, Hem, Sketched Bend, Jog
Forms and details	Cross Break, Tab and Slot, Forming Tools, Gusset
Corners	Corner Relief, Closed Corner
Flat	Unbend, Bend, Flatten
Checking and output	Sheet Metal Check, Export DXF, Export All DXF, DXF Settings

15.1 Base Flange

A closed sketch (holes allowed) becomes a sheet of the given **thickness**. The **bend radius** and **K-factor** set here become the part's defaults.

15.2 Gauge tables

Base Flange, Sheet Metal Extrude and Transition / Reducer have a **Gauge** list: mild steel, steel gauges, stainless 304, and aluminium 5052 and 6061-T6. Picking a gauge sets the thickness, inside bend radius and K-factor. **Save...** keeps your own values as a gauge.

15.3 Sheet Metal Extrude

Extrudes an open sketch of lines and arcs into a sheet metal part. Arcs in the sketch are bends of their own radius; sharp corners get a bend of the inside radius set in the panel. Material goes on one side, the other side or both sides of the sketch; depth one way, reversed or mid-plane. A circle on its own is rolled into a cylinder, leaving a seam gap. Flanges, hems, Flatten and DXF export work on it.

15.4 Lofted Bend

A sheet metal part between two open profile sketches (each one chain of lines and arcs), for example a round-to-square transition, hopper or duct piece. Points are matched along both profiles by length, keeping every corner of either profile.

- **Method**: **Bent** makes flat facets folded along straight lines, each fold a bend line in the flat pattern and DXF; set the number of divisions. **Formed** makes a smooth sheet (rolled or pressed), with the set number of forming lines along the profiles.
- The flat pattern is developed at the neutral layer (K-factor) and works with Flatten, DXF export and flat pattern drawings. **Material on the other side** flips the thickness.
- A cut through a lofted bend is carried into its flat pattern, and the cut edges are square to the sheet, as a laser-cut blank formed to shape would be. This makes trimmed conical parts possible: loft the cone between two arcs, then cut it to the outline with a revolved cut.
- Edge flanges, corner reliefs and closed corners can't be added to a lofted bend; its shape comes from the profiles.

15.5 Transition / Reducer

Cones, concentric and eccentric reducers, square-, rectangle-, triangle-, pentagon- and hexagon-to-round transitions (either way up) and polygon frustums.

- Sizes, offset and rotation are set in the panel; sizes can be inside, outside or mean. A sketch line from bottom centre to top centre places the part and sets its height; change the line and the part follows. Started with no sketch, it asks for a plane and opens the Line tool.
- The part is made from a number of equal pieces, each a sheet metal body with its own flat pattern: a cone from one piece or more, a flat-sided transition from two or more, with seams at the middle of the flat sides or at the corners.
- **Formed** (rolled, with a set number of forming lines per corner or round the whole part) or **bent** (bump formed with a set number of bends).

15.6 Edge Flange

Click straight edges on the top or bottom face.

Setting	Options
Length	Measured to the outer virtual sharp, inner virtual sharp or tangent
Angle	Flange angle
Position	Material inside, material outside, bend outside, plus an offset; flip
Bend radius	Its own radius, or the default
Auto relief	Rectangular, obround or tear relief where a bend ends next to material
Corners	Flanges picked together that meet at a corner are closed with a gap (default 0.1 mm), or left open. Corner type Overlap, Underlap or Butt; **Swap** reverses which covers; Bend region V cutout or Closed.

15.7 Miter Flange

A flange with a profile of up to four segments along a chain of edges. Each segment has a length (to the outside of the bends) and an angle; a negative angle folds back the other way, so 90 / 90 makes a C and 90 / −90 a Z. The corners are mitred (closed, with a gap) at every level. **First bend** sets where the material sits on the edge, as for an edge flange.

15.8 Hem

Folds a straight edge back on itself. Hems meeting at a corner are mitred.

Setting	Options
Type	**Closed** (pressed flat), **Open** (with a gap), **Teardrop** (its own angle, default 225°) or **Rolled** (a curl)
Length	From the outside of the fold
Gap, inside radius, angle	As the type needs
Position	**Inside** (the fold at the edge) or **Outside** (added beyond the edge); flip direction

15.9 Sketched Bend

Draw lines on the top or bottom face of a flat part of the sheet; the sheet is bent along each one. Set the angle, the radius (or the part's), whether the bend is centred on the line or starts or ends at it, the fold direction and which side stays put. The flat pattern stays the plate it was.

15.10 Jog

One sketch line on a face makes two opposite bends that step the sheet over by a height. The height is measured face to face, outside to outside or inside to inside; the jog says when the height is too small for the bends.

15.11 Cross Break

Click flat faces to mark an X of shallow stiffening folds. It shows in Flatten, in the DXF (on the Forming marks layer, with a note) and in flat pattern drawings.

15.12 Tab and Slot

Click the edge of one sheet metal part where it sits on another, then the face of the other part. Tabs are added along the edge and matching slots, with clearance all round, are cut through the other part, so the parts locate each other for welding.

- Settings: number of tabs, tab length, tab height (empty: through the other part), margin at each end (empty: evenly spread) and clearance.
- Both flat patterns and DXFs get the tabs and slots.

15.13 Forming Tools

Forming Tools presses a form into the sheet. Place forms at clicked points, or exactly at the points and circle centres of a sketch on the face (for a louver, its lines). They are pressed toward the face they are placed on, or the other way.

Form	Settings
Dimple	Diameter, depth
Extruded (flanged) hole	Hole diameter inside the collar, collar height, bend radius
Louver	Width back from the slit, opening height, hood side; drawn as a line on the face (the slit)

The flat pattern gets a centre mark and a note for each form, an extruded hole's pre-hole (the collar laid flat) and a louver's slit as a cut. Forms stay with the part through later sheet metal features and are left off, rather than breaking the part, if one can't be made.

15.14 Gusset

Click a bend (its round face) to press a stiffening dart across it at that point. Set the size of each leg from the inside corner and the width across the bend; tick **Pressed down** to press it the other way. The flat pattern marks it. (For gusset plates between structural members, see the Fast Structure chapter.)

15.15 Corner Relief

Adds a rectangular, circular or tear relief at every corner where two bends meet.

15.16 Closed Corner

Closes the corners between 90° flanges: one flange covers the other's end, with a gap.

- **Corner type**: Overlap, Underlap or Butt (mitred, gap across the mitre, bends included).
- **Which corners**: click a face of a flange next to a corner and the matching flange is found automatically; the clicked flange is the one that overlaps or tucks under. With no corners picked, every corner of the body is closed.
- **Swap** reverses which flange covers.
- **Bend region**: **V cutout** (default) runs each bend into the corner as it would form from the flat, leaving a small teardrop-shaped opening with a relief at the corner point. **Closed** keeps the bends meeting on the diagonal.
- The flat pattern matches the folded part at closed corners, including each bend's run-on, the relief and the corner round.

15.17 Unbend and Bend

Unbend keeps the picked face in place and flattens the rest. Features after it (such as cuts across bends) are made in the flat. **Bend** folds it back to where it was. Both appear in the tree.

15.18 Cuts and other features on sheet metal

Cuts on sheet metal (extruded cut and so on) become holes in the flat pattern, so the flat pattern always matches. Holes (the Hole feature) work on sheet metal too. Other features that change a sheet metal body are allowed, but the flat pattern can't follow them; this is shown in Body Properties and the DXF preview.

15.19 Flatten

Click **Flatten** on the tab, or right-click the body ▸ **View Flat Pattern**. The flat pattern is shown on its own; **Esc** or Flatten again restores the view as it was. While it is shown, **Export DXF…** on the flat pattern banner (or right-click the view) exports it straight away.

15.20 Convert to Sheet Metal

Turns an ordinary body into sheet metal. Pick a flat face of a thin-walled body (for example a STEP export of a sheet metal part, or a shelled box), or of a knitted surface of flat faces (enter the thickness).

- RazorCAD follows that side of the sheet across every smooth edge, through flat faces and cylindrical bend faces, so the whole body is found. Each bend keeps its own radius and angle.
- Bends that run straight into each other with no flat between (rolled shapes such as a fan scroll) are joined automatically. Cut-outs in bends are kept and holes stay true circles in the DXF.
- Sharp-cornered bodies (shelled boxes) get bends of the chosen radius at every edge between two flat faces.
- The panel shows what was found in the body, for example "thickness 3 mm · bend radii R300, R227, R13 (inside, 3 bends)" or "4 sharp edges".
- **Use the body's thickness** and **Use the body's bend radii** are on by default. Untick either to set your own value (pre-filled with the detected one).
- After converting, the sheet's volume is compared with the body's. The feature note warns when they differ by more than 0.5 % (for example a nut or motor part that isn't sheet metal).

- A **curved surface** that is curved one way only (a cylinder, or a profile swept straight) is converted into a rolled sheet: its section is followed by smoothly joined arcs, each a bend of its own radius. **Arc fit tolerance** (default 0.05 mm) sets how closely it follows the surface. **Thickness** goes behind the surface, in front of it, or both sides.
- **Trim where the sheet would run into itself** (on by default) trims a rolled sheet back where its two ends would meet, leaving a small clearance.

15.21 Sheet Metal Check

Sheet Metal Check lists every bend of the sheet metal body: direction, angle, inside radius, length and the straight flange on each side. It warns about flanges shorter than 3 × thickness past their bend, and holes closer than 2 × thickness + radius to a bend.

15.22 Sheet metal Body Properties

Body Properties for a sheet metal body shows thickness, bend radius and K-factor, number of bends, overall flat size, flat area, mass and **quantity**, plus:

- **Cut length**: the total length of the flat outline plus every hole and cut-out, and the number of **pierces** (closed contours), for laser-cutting estimates. Also shown in the DXF part information and under flat pattern drawing views.
- **Flat pattern seen from**: top face or bottom face. The bottom face mirrors the pattern and swaps UP / DOWN bend lines and notes. Flatten, DXF export and drawing views all follow it.

15.23 Exporting flat patterns as DXF

- Right-click a sheet metal body (tree or view) ▸ **Export Flat Pattern as DXF**, or **Export DXF** on the tab.
- **Export All DXF**: choose a target folder; one file per body, named after the body. Identical bodies give one file with the quantity multiplied by the count.
- Each file is shown in a **preview window** with its layer colours and line types, which can be changed there. Buttons: **Export**, **Skip This One** and **Abort Export**.
- In the preview, zoom with the wheel, pan by dragging and double-click or **Fit** to see it all. Click lines to select them (**`Ctrl`** or **`Shift`** adds, **`Shift`**+drag for a box, **`Ctrl+A`** for all), then **Delete** them, give them their own **Line type**, or move them to another **Layer** (for example make a line an etch line). **Undo** and **Restore All** put them back. These edits apply to that one export.
- Options: bend lines, bend notes, bend region (hidden) lines, etch lines and part information.

15.24 DXF Settings

DXF Settings on the Sheet Metal tab sets the defaults for every export: layer mapping (colours and line types) and options. Defaults: **up bends red**, **down bends blue**, **etch lines yellow dashed**. **Reset to Defaults** restores them.

15.25 Not yet available

Etch lines drawn from sketches (the layer exists).

16 Fast Structure

The **Fast Structure** tab builds welded frames (weldments) from standard steel profiles: Structural Member, Trim/Extend, End Cap, Gusset, Weld Bead and Cut List. It also has a **3D Sketch** button, since frames are often laid out in 3D.

16.1 Structural Member

Pick a profile and a sketch or 3D sketch: every line and arc of it becomes a member, each its own body named after its profile. Add more sketches to the **Path** list for a frame in 3D, for example the top of a table and its legs; the joints between them are treated too. Arcs, and the bends of a 3D sketch with a bend radius, become bent members.

Profile	Types
Hollow sections	Square (SHS), rectangular (RHS), circular (CHS / pipe)
Angles	Equal and unequal angle
Channels and beams	Channel (PFC / UPN); I-beam (UB, UC, IPE, HEA)
Bars	Flat bar, round bar

Sizes come from AS/NZS or EN/ISO tables, or enter your own dimensions.

- **Corners** where two members meet end to end: **Mitre**, **Butt** (first or second member trimmed) or **None** (members meet at the sketch points). **Trimmed ends**: square cut on the face, or coped to the other member's outside.
- A member ending on the middle of another (a T-joint), or under a corner (a table leg), is trimmed to it, square on its face. Pipes and round bars are always coped (fish-mouthed).
- **Alignment**: which point of the profile the sketch line runs through (centre, corners or edges), a profile rotation, and mirror.

16.2 Trim/Extend

Click the members to trim on the model, then the boundary: a plane or another body. Each member is trimmed (or, with **Extend members that stop short of it**, extended) at its end nearer the boundary, either cut **square** on the boundary's face or **coped** to the boundary body.

16.3 End Cap

Click the open end faces of tubes or pipes. A plate of the set thickness is added **on the end** (outside) or **in the end** (flush).

16.4 Gusset

Click two flat faces that meet at a corner, one on each member. A triangular plate fills the corner. Set the two leg lengths, the thickness, a **corner chamfer** for weld clearance, and where the plate sits along the corner (centred or to one side, plus an offset).

End caps and gussets are bodies of their own, named by size for the cut list.

16.5 Weld Bead

Set the weld size and click the bodies to weld (none: every weldment member). A bead runs along every line where two of them touch; each joint is its own bead body.

16.6 Cut List

Cut List lists identical members together, with their profile, length, end angles, quantity, total length and mass (from the bodies' materials), then plates and welds (with the total weld length). It also gives the total length of each profile. **Export CSV…** saves it for a spreadsheet. A drawing can show it as a **Cut List Table** (Drawings chapter).

17 Configurations

A configuration is a variant of a part or assembly: other values for the global variables and other features suppressed in a part, other part configurations and components shown in an assembly.

17.1 The Configurations tab

The **Configurations** tab sits next to the Feature tree and the Property panel at the top of the left panel. It lists **Default** and every configuration, the one shown ticked, each with a summary of what it changes.

- **Double-click** a configuration to show it. The tree shows its name next to the part.
- **+ Add Configuration** makes a new one starting from the one shown.
- Right-click for **Show Configuration**, **Rename**, **Copy Configuration**, **Properties (Design Table)** and **Delete**.

17.2 What a part configuration changes

- **Variable values**: each configuration can give the global variables (Tools ▸ Equations) other values.
- **Suppressed features**: while a configuration is shown, **Suppress / Unsuppress** in the tree changes that configuration only. The tree greys out what it suppresses.

17.3 Design table

Design Table… in the Configurations tab (or right-click ▸ **Properties (Design Table)**, or **Configurations** on the Evaluate tab) edits all the configurations in one table. The table can be copied to and pasted from a spreadsheet as CSV.

17.4 Configurations in assemblies

Assemblies have the same tab. While an assembly configuration is shown, these change that configuration only: a component's part configuration, **Hide / Show**, **Suppress** (a suppressed component is left out, with its mates) and a mate's Suppress or value. See the Assemblies chapter for how components use part configurations.

18 Assemblies

An assembly places parts and other assemblies as **components** and holds them together with **mates**. Assemblies are saved as **.rasm** files and open in a tab of their own.

18.1 Starting an assembly

File ▸ New Assembly, or **New Assembly** in the start window. Every assembly has the standard **Front, Top and Right Planes** and the **Origin** at the top of its tree. They show in the view as a part's do (the eye in the tree, **View ▸ Show Planes**), and mates can use them; they stay where they are.

The **Assembly** tab of the ribbon holds Insert Components, Mate, Fix, Float, Hide Component, Suppress, Open Part, Exploded View, Explode / Collapse, Bill of Materials and Zoom to Fit.

18.2 Inserting components

Insert Components lists the parts and assemblies open in other tabs, or **Browse for a Part or Assembly…** takes a .rcad or .rasm file.

- The first component is **fixed** at the origin; the others are placed beside what is there.
- If the part has configurations, you choose which one to use. Right-click a component later to switch it to another configuration. The same part can be in the assembly several times in different configurations.
- The assembly keeps a copy of each part, refreshed from its tab or file whenever you come back to the assembly, so edit a part and the assembly follows.
- Several instances of one part share it: it is built once and drawn once.

18.3 Mates

Click **Mate** (`M`), then two faces, edges or vertices on different components (or a standard plane), choose the type and click ✓. The panel stays open for the next mate; `Esc` closes it.

Mate	Holds
Coincident	Faces, edges or points together
Concentric	Axes in line
Parallel / **Perpendicular**	Directions
Tangent	A face touching a round face
Distance / **Angle**	At a set value
Lock	One component fixed to another

- Flat faces count as planes; cylinders, cones and circular edges as axes; spheres and vertices as points.
- Each mate is shown applied while you choose. **Flip** swaps Aligned / Anti-Aligned (or the side for Distance and Tangent). When the alignment chosen from how the faces sit can't be reached, the other one is used.
- A mate that conflicts with the others is refused with a message.

18.4 Moving components

Left-drag a component to move it, **Alt**+drag to turn it. It moves only as its mates allow. The view turns, pans and zooms as in a part (see Navigating the 3D View).

Selecting works as in a part and follows the **selection filter**: with no filter a click picks the vertex, edge, face or plane under the cursor; with the Bodies filter it picks whole components. **Ctrl**+click adds or takes away. Right-click a face, edge or vertex for **Mate…** and the component's commands; whatever is selected goes into the mate. Dragging a component moves it, whatever is selected.

18.5 The assembly tree

- Each component shows **(f)** when fixed or **(-)** while still free to move, its configuration, and an eye to hide or show it.
- The **Mates** folder lists the mates; right-click one for **Edit**, **Flip**, **Suppress** or **Delete**.
- Right-click a component for **Mate…**, **Open Part** (or double-click it), **Fix / Float**, **Hide / Show**, **Suppress / Unsuppress**, **Insert Another Instance**, **Rename** and **Delete**.
- Undo and redo work in the assembly.

18.6 Sub-assemblies

An assembly can hold other assemblies as components. **Insert Components ▸ Browse for a Part or Assembly…** takes a .rasm file, and STEP assemblies keep their sub-assemblies when they open as an assembly. An assembly can't go inside itself.

- A sub-assembly is kept whole, with its own parts, placements and configurations, and moves, hides, suppresses and explodes as one. Clicking any of its parts picks it.
- Mates can use the faces, edges and vertices of the parts inside it.
- In the tree, ▶ opens a sub-assembly to show its components. Double-click one, or right-click the sub-assembly ▸ **Open Sub-assembly**, to change it in its own tab; the changes come back to the assembly.

18.7 Exploded views

Exploded View on the ribbon (or **Insert ▸ Exploded View…**) spreads components apart for display; the model and its mates stay put.

1. Click components to pick them (click again to drop one).
2. Pick a direction: ±X, ±Y, ±Z, or along a face's normal.
3. Drag the picked items along the direction, or type the distance, then click **Add Step**. **Spread Out** spaces the picked items out along the direction, one step each.
4. Click a step to change it, ✕ to delete it. **Explode lines** (dashed, from each item's place to where it is shown) can be switched off.

The tree's **Exploded Views** folder lists them. Double-click one to explode or collapse it (animated); right-click for Explode / Collapse, Animate, Edit, Rename and Delete. The **Explode / Collapse** button on the ribbon does the same for the last one. Dragging a component, adding a mate or editing collapses the view first. Exploded views are saved with the file.

18.8 Bill of Materials

Bill of Materials lists the parts with their quantities and materials: one row per part and configuration used, with suppressed components left out. It is indented: a sub-assembly's parts are listed under it (1.1,

1.2…), with the quantities in one of it. **Copy as CSV** or **Save CSV…** takes it to a spreadsheet.

18.9 STEP assemblies

Opening a STEP file that holds an assembly asks whether to open it **as an assembly** or **as one part** (see Importing in the Working with Files chapter). As an assembly, each part of the file becomes a part, each sub-assembly a sub-assembly, and each component is placed, fixed, where the file places it, with the names and colours from the file.

18.10 Saving parts as separate files

When an assembly holds parts or sub-assemblies that are only inside it (as a STEP assembly does when it opens), Save asks how to save them:

- **Save Parts as Separate Files** saves each part as a .rcad file and each sub-assembly as a .rasm file in the assembly's folder, named after them. The assembly (and each sub-assembly) then uses those files, so each part can be opened and changed on its own and the change shows in the assembly. A part used in several places is one file. If files of those names are already there, you are asked before they are replaced.
- **Save as One File** keeps everything inside the assembly file. You are not asked again, except on Save As.

18.11 Big assemblies

Each part an assembly builds is kept in the model cache (`%LOCALAPPDATA%\RazorCAD\modelcache`). Opening the assembly again shows unchanged parts straight from there, and the kernel only builds parts that changed, so a large layout reopens in seconds. Each part is sent to the graphics card once and placed at every component that uses it, so moving, hiding or exploding components is instant.

18.12 Not yet available

Interference check of assembly components (Interference Check works on the bodies of a part) and STEP export of a whole assembly.

19 Drawings

RazorCAD makes 2D engineering drawings of parts and assemblies, saved as **.rdrw** files and opened in their own tab.

19.1 Starting a drawing

- **Start window ▸ New Drawing**, or **File ▸ New Drawing**, opens the set-up dialog (below) and then an empty sheet.
- **File ▸ Make Drawing from Part** (in a part), or **Make Drawing from Assembly** (in an assembly): after the set-up dialog, places the standard three views at a scale that fits, clear of the title block.
- A drawing can show assemblies as well as parts: every component is drawn where it sits, through its sub-assemblies, in the configurations shown; suppressed and hidden components are left out. Big assemblies (more than about 200 solids) are drawn fast from the parts' display meshes: lines, circles and arcs as exact as the 3D view's, and where an edge turns hidden to about two thousandths of the view's size. A site layout of thousands of solids then takes seconds a view instead of many minutes. In a view's properties, **Hidden lines** chooses Automatic, Exact or Fast; section and flat-pattern views are always exact.

Drawings are opened, saved and listed in the recent files like parts.

19.2 The sheet and its template

The default template is A3 with an ISO 5457 frame (zones and centring marks) and an ISO 7200-style title block: title, drawing number, material, scale, size, units, projection symbol, drawn / checked / approved, date, mass, company, revision, sheet n of m and general tolerances. Sheet sizes A0 to A4, landscape or portrait.

19.3 Drawing templates

A **drawing template** (.rdrt file) keeps a drawing's set-up: its sheets (size, orientation, scale, projection), the drawing settings (text height, decimals…), the title block's standing entries (company, drawn / checked / approved by, finish, tolerances, revision) and notes placed on the sheet. Views, parts and part-specific entries (title, drawing number, material, mass, date) are not kept.

- **File ▸ Save as Template…** in a drawing saves one, by default to `Documents\RazorCAD\Templates`. Opening a .rdrt file edits the template in its own tab.
- **The New Drawing dialog** lists your templates (from the Templates folder, or **Browse…** for one anywhere) and the standard ones: A4 portrait and landscape, A3, A2, A1 and A0 landscape, and A3 and A4 in third angle projection. Set the drawing name, title, drawing number, drawn by and company, and the first sheet's size, orientation, scale and projection. Tick **Use this template as my default** to have it chosen first next time. Double-click a template to create the drawing at once.
- **Change Template…** (File menu, Sheet tab, or right-click the drawing in the tree) switches an existing drawing to another template. Views, dimensions and annotations stay; company, tolerance and finish come from the new template, and the old template's notes are replaced. Undo goes back.
- **Company sheet formats**: a template can carry its own border, title block and logo in place of the standard ISO one, used on sheets of its size and orientation. Its text shows the drawing's properties, and can be set in a font installed on the PC; PDF exports embed the font. Drawing Properties lists the title block's own fields.

Changing the projection (Change Template or Sheet Properties) lays the views out again for the new projection, so a top view stays a top view.

19.4 Working on the sheet

Action	How
Zoom	Mouse wheel
Pan	Middle or right drag, or left drag on empty paper
Fit the sheet	**F**
Move views and annotations	Drag them
Edit	Double-click a view for its properties, a dimension or note to edit it
Menus	Right-click
Delete	**Del**
Undo / Redo	**Ctrl+Z** / **Ctrl+Y**

19.5 View Layout tab

Command	What it does
Model View	Pick an open part or assembly, or browse for its file (.rcad or .rasm), pick the orientation (front, back, top, bottom, left, right, isometric, dimetric, trimetric or flat pattern) and click on the sheet. Then move around it to place projected views.
Standard Views	Three views laid out for first- or third-angle projection
Projected View	A view aligned with its parent
Flat Pattern	A sheet metal body unfolded (see below)
Section View	Horizontal or vertical cutting line, hatched cut faces, A-A labels and arrows, flip
Detail View	A circle with a letter, 2:1 by default
Auxiliary View	Looks square at an inclined edge, so it shows its true shape
Break	Leaves out the middle of a long part; dimensions across the break keep the true length
Crop View	Shows only the area you draw round
Update Views	Re-reads the parts and redraws the views

A broken or cropped view can be put back to whole from its right-click menu. Section views hatch each body or component with its own angle and spacing.

Views are hidden-line projections computed by the geometry kernel.

19.6 View properties

Click a view to open its panel:

- Orientation, changeable after placing (also to and from Flat Pattern; projected and section views made from it are removed when it becomes a flat pattern, and **Ctrl+Z** brings them back).
- Display style: hidden lines removed, hidden lines visible, or wireframe; tangent edges on or off.
- Sheet scale or a custom scale; labels.

- **Rotate View**: −90°, +90°, 180° and 0° buttons and an angle box (degrees, counter-clockwise). Section lines, detail circles and annotations turn with the view.

Aligned views slide only along their alignment. Deleting a view deletes the views made from it.

19.7 Flat pattern views

A flat pattern view shows a sheet metal body unfolded with its outline and holes, bend lines (chain lines) with notes (UP / DOWN, angle, radius), optional bend region lines, and its flat size, thickness, number of bends and cut length underneath. It can be dimensioned (bend lines too) and exported to PDF. If the part has several sheet metal bodies, pick the body in the view properties. Instead of notes, the view can show a **bend table**: the bend lines numbered B1, B2… and a table of direction, angle and inside radius under the view. Forms, cross breaks and tabs and slots show as in the DXF.

19.8 Annotation tab

Command	How
Dimension	Click a line for its length, a circle for its diameter (an arc: radius), two points, lines or circles for the distance between them (two lines at an angle: the angle), then click where the text goes. Horizontal, vertical or aligned depends on where you place it. Prefix, suffix, tolerance and override text can be set.
Note	Click on the sheet; click an edge first to add a leader
Baseline Dimension	Click the base edge or point, then each point or edge to dimension from it; the dimensions stack one above the other
Ordinate Dimension	Click the origin, then where the values line up, then the points to dimension
Chamfer Dimension	Click the chamfer edge and an edge next to it, then place the text ("3 × 45°")
Hole Callout	Click a hole. The callout gives its size and depth from the Hole feature (counterbore, countersink, tapped thread and tap drill) and how many of the same holes the view shows, for example "4× Ø6.6 THRU ALL / CBORE Ø11 DEEP 6.5".
Datum Feature	A datum letter in a box, with a triangle on the feature
Geometric Tolerance	A feature control frame: one of the 14 characteristic symbols, tolerance with Ø zone and material condition, and up to three datums
Surface Finish	A surface finish symbol with its roughness
Weld Symbol	Fillet, square, V, bevel or plug weld on a reference line with an arrow to the joint; arrow, other or both sides; all-around, field weld and tail
Center Mark	Click circles or arcs
Centerline	Click two lines (or two points); the centerline runs between them
Revision Table	Rows edited in one dialog; the latest revision goes into the title block
Cut List Table	The cut list of the part in the selected view (see the Fast Structure chapter); it updates when the part changes

Datum features, geometric tolerances, surface finish and weld symbols can each have a leader. Right-click one ▸ **Edit…** to change it.

19.9 Sheet tab

- **Sheet Properties**: size, orientation, scale, first or third angle.

- **Title Block** (Drawing Properties): title block fields, text height and decimals.
- **Change Template**: see Drawing templates above.
- **Add Sheet**: drawings can have several sheets, shown as tabs under the sheet.
- **Export PDF**: vector PDF, one page per sheet.
- **Export DXF**: the current sheet as a DXF file, with layers for visible, thin, hidden and centre lines, text and hatching. A sheet format's logo is left out.

19.10 Views follow the part

Views and dimensions follow changes to the part or assembly, whether it is open in another tab or read from its file. A copy of the part is kept in the drawing so it still opens if the part file is missing.

19.11 Not yet available

Balloons and bill-of-materials tables. DWG export: DWG is a closed format, so drawings are exported as DXF, which practically every CAD program opens.

20 Evaluate and Tools

20.1 Measure

Evaluate ▸ Measure, **Tools ▸ Measure...** or the view's right-click menu. Click vertices, edges, faces, planes or sketch lines; with two items picked it shows the distance between them plus dX, dY and dZ. Use the body filter (**B**) to pick whole bodies.

20.2 Mass Properties

Evaluate ▸ Mass Properties shows mass, volume, surface area and centre of mass. It adds up every body with its own density and lists the bodies. A body with no material uses 1.00 g/cm³.

20.3 Interference Check

Evaluate ▸ Interference Check finds where the solid bodies of a part overlap. It lists each pair with its overlap volume and shows the overlaps in red.

20.4 Surface analysis

These displays colour the model to show the quality of its faces. Click the button again to turn its display off.

Command	Shows
Zebra Stripes	Reflected stripes on every body. Across a smooth join the stripes run on without a kink.
Curvature	Bodies coloured from blue (flat) through green and yellow to red (tightest). The status bar gives the radius red stands for.
Draft Analysis	Choose a pull direction (an axis or a selected face's normal) and the draft needed. Faces turn green with enough draft, yellow where they need more, and red with draft the other way.

20.5 Configurations

Evaluate ▸ Configurations opens the design table (see the Configurations chapter).

20.6 Equations and global variables

Tools ▸ Equations... holds global variables that can be used in any dimension or feature value. Example: `width = 120`, then set a depth to `width/4`. Each row shows the value it evaluates to; invalid or duplicate names are flagged.

Every value field accepts:

Item	Examples
Operators	`+ - * / ^ %` and parentheses
Variables	Any name from Equations
Trigonometry (degrees)	`sin cos tan asin acos atan`
Other functions	`sqrt abs min max round floor ceil pow`

Item	Examples
Constants	`pi`

A value may start with `=`. Variable names start with a letter or underscore and can't be a function name.

20.7 Edit Material

Tools ▸ Edit Material… (or the Material item in the tree) sets the part material. See Appendix B for the list.

21 Modelling with Claude

RazorCAD can build features from a plain-language description using Anthropic's Claude.

21.1 Setting up

Open **Tools** ▸ **Claude Settings...** (or the Claude tab ▸ **Claude Settings**) and choose how RazorCAD reaches Claude:

- **My Claude subscription (Claude Code on this PC)**: each request goes through the Claude Code command-line program installed on this PC, signed in to your own Claude Pro or Max plan. It counts toward that plan's usage limits, with no separate bill. The dialog shows where Claude Code was found, its version and whether it is signed in; **Sign In...** starts the sign-in, which you finish in the browser, and **Check Again** reads the status again. Claude Code runs with no tools, settings or memory, from an empty folder, so it sees only RazorCAD's modelling guide and your request.
- **An Anthropic API key**: enter your own key (create one at console.anthropic.com ▸ API Keys; requests are billed to that account). The key is stored on this PC in `%APPDATA%\RazorCAD\claude.json` and is only sent to api.anthropic.com. **Remove Key** deletes it.

Then choose a model: Opus, Sonnet or Haiku.

21.2 Ask Claude panel

Open it with the **Claude** tab ▸ **Ask Claude**, or **Tools** ▸ **Ask Claude...**.

- Describe a part or a change. Claude adds the features to the tree; each answer is **one undo step**.
- If the features don't build, Claude is told the error and corrects it, up to two more tries.
- Claude sees the feature list and body sizes of the part, not the picture.
- **New Conversation** (or **New chat** in the panel) starts fresh; the part stays as it is.
- The model can be switched in the panel.

21.3 RazorCAD Part Builder on claude.ai

A companion page on claude.ai lets you describe a part, see the feature list with sketch previews, and download it as a .rcad file to open in RazorCAD. It uses the same modelling guide as the in-app panel.

22 Troubleshooting

22.1 Crash reports

RazorCAD writes a crash report whenever something goes wrong, to `%APPDATA%\RazorCAD\crashes\<date_time>_<kind>\`. Open the folder with **Help ▸ Open Crash Reports Folder**. The start window mentions new reports. The newest 30 are kept.

File	Contents
`report.json`	The kind of failure, the error message and stack trace, what RazorCAD was doing (mode, Property panel, sketch tool, selection, feature errors) and system information
`session.log`	The recent actions of the session, newest last
`part.rcad`	The part that was open, so the problem can be reproduced

Kinds of report: `ui-error`, `kernel-error`, `kernel-worker-crash`, `python-exception`, `native-crash` and `unexpected-shutdown` (a session that ended without closing normally, found at the next launch).

22.2 Common problems

Problem	What to do
A fillet or projected edge loses its edge after a big change	Edge references are found again by geometry, not by true topological naming. Edit the feature and re-pick the edge.
A sketch on a model face doesn't follow the face	Sketches on faces are stored as fixed planes. Use a reference plane driven by an equation when that matters.
Project Edges can't see a sketch	It only sees visible sketches earlier in the history. Show the sketch first.
An STL body can't be filleted	STL bodies with thousands of faces can't be filleted or chamfered and don't offer pickable edges.
A fillet won't blend across two touching bodies	Combine the bodies first.
Open fails with a file filter error	Fixed in 0.9.0-beta. Update RazorCAD.
Unsaved work after a crash	Reopen the part and accept the auto-save recovery.

22.3 Settings and data files

Location	Contents
`%APPDATA%\RazorCAD\recent.json`	Recent files list
`%APPDATA%\RazorCAD\claude.json`	Claude settings, API key and model
`%APPDATA%\RazorCAD\prefs.json`	Toolbar layout, icon size and other preferences
`%APPDATA%\RazorCAD\crashes\`	Crash reports
`%LOCALAPPDATA%\RazorCAD\modelcache\`	Built models of parts and assemblies, for fast opening (up to 1 GB; safe to delete)
`Documents\RazorCAD\Templates\`	Your drawing templates
`Name.autosave.rcad`	Auto-save, next to the part file

23 For Developers

23.1 Rebuilding the executable

```
python -m pip install pywebview pyinstaller pillow
build.bat
```

`build.bat` bundles `app/` (including the 23 MB OpenCascade WebAssembly kernel) and `razorcad.py` into `RazorCAD.exe` with PyInstaller. If RazorCAD is open, the running copy is renamed to `RazorCAD.old.exe` and the new build is used from the next launch.

`installer\build_installer.bat` builds `RazorCAD.exe` and then `RazorCAD-Setup.exe`. Set the version in `version.txt` first.

23.2 Working on the UI

Run `python serve.py` and open http://localhost:8765/ in a browser. In that mode files are downloaded instead of saved through dialogs. Set the `RAZORCAD_DATA_DIR` environment variable to keep a test run's recent-files list separate from your own.

23.3 Project layout

Path	Purpose
`RazorCAD.exe`	The application (built by build.bat)
`razorcad.py`	Desktop shell: native window, file dialogs, auto-save (pywebview)
`app/js/main.js`	Application controller: documents, commands, selection, files
`app/js/ui.js`	Menus, Command ribbon, Feature tree, Property panel, dialogs
`app/js/model.js`	Feature definitions, undo history, materials, sample part
`app/js/kernel.worker.js`	Geometry kernel worker: replays the history with OpenCascade
`app/js/sketcher.js`, `sketchmodel.js`, `solver.js`	Sketch mode, sketch data and the constraint solver
`app/js/sketch3d.js`, `sketch3dmodel.js`	3D sketches
`app/js/sheetmetal.js`	Sheet metal features and flat patterns
`app/js/weldments.js`, `profiles.js`	Fast Structure members and the profile tables
`app/js/assembly.js`, `asmview.js`, `explode.js`	Assemblies, their display and exploded views
`app/js/drawing*.js`, `drawtemplates.js`, `pdf.js`, `dxf.js`	Drawings, templates, sheet formats, PDF and DXF export
`app/js/claudespec.js`	Modelling guide used by Ask Claude
`Manual/`	This manual: source, build script and change tracking

Appendix A: Keyboard and Mouse Reference

General

Action	Keys
New / Open / Save	**Ctrl+N** / **Ctrl+O** / **Ctrl+S**
Save As	**Ctrl+Shift+S**
Import STEP / STL	**Ctrl+I**
Undo / Redo	**Ctrl+Z** / **Ctrl+Y**
Rebuild	**Ctrl+B**
Delete feature or entity	**Del**
Accept / cancel Property panel	**Enter** / **Esc**
Edit a feature	Double-click it in the tree

View

Action	Keys
Zoom to fit	**F**
Front, Back, Left, Right, Top, Bottom, Isometric	**Ctrl+1** … **Ctrl+7**
Normal To	**Ctrl+8**
Rotate in steps	Arrows 15°, **Shift** + arrows 90°
Pan in steps	**Ctrl** + arrows
Roll	**Alt** + ← / →
Hide body under cursor / show hidden body under cursor	**Tab** / **Shift+Tab**

Selection filters

Filter	Key
Faces / Edges / Vertices / Bodies	**X** / **E** / **V** / **B**
Clear filters	**F5**

Sketch

Tool	Key
Select	**S**
Line	**L**
Rectangle	**R**
Circle	**C**
Arc	**A**

Tool	Key
Spline	**N**
Polygon	**P**
Trim	**T**
Dimension	**D**
Project Edges	**E**
Offset	**O**
Construction Geometry	**Q**
Delete	**Del**
End tool, then exit sketch	**Esc**
Tool wheel	Hold the right button (or drag with it)

3D sketch

Action	Keys
Next drawing plane (XY → YZ → ZX), or back from a picked plane	**Tab**
Draw off the X/Y/Z directions	Hold **Ctrl** or **Shift**
Draw on a plane or flat face	**Ctrl**+click it
Length dimension: true length → ΔX → ΔY → ΔZ	**Tab** while placing
Tools	Same keys as 2D sketches: **S**, **L**, **C**, **A**, **N**, **P**, **D**, **T**, **E**, **Q**

Assembly

Action	Keys
Mate	**M**
Move a component / turn it	Left-drag / **Alt**+drag
Zoom to fit	**F**

Appendix B: Materials

Material	Density (g/cm³)
Plain Carbon Steel	7.80
AISI 304 Stainless	8.00
6061 Aluminum Alloy	2.70
Brass	8.50
Copper	8.96
Ti-6Al-4V	4.43
ABS Plastic	1.04
PLA Plastic	1.24
Nylon 6/6	1.14

A body or part with no material uses 1.00 g/cm³ for mass calculations.

Appendix C: Manual Revision History

Date	Software version	Changes
2026-10-20	0.12.0-beta	Brought up to date with the software. New chapters: 3D Sketches, Airfoils, Fast Structure, Configurations and Assemblies. Sheet Metal adds Sheet Metal Extrude, Lofted Bend, Transition / Reducer, Miter Flange, Hem, Sketched Bend, Jog, Cross Break, Tab and Slot, Forming Tools, Gusset, gauge tables and Sheet Metal Check. Drawings add templates, auxiliary, break and crop views, baseline, ordinate and chamfer dimensions, hole callouts, GD&T, surface finish and weld symbols, revision and cut list tables, bend tables and DXF export. Surfaces add lofted, swept, filled, boundary and ruled surfaces, trim, extend, untrim, delete and replace face, split line, projected and composite curves, flatten surface, convert to solid and repair mesh. Also: new sketch tools, Hole, Draft, Rib, Dome, Move Face, Scale, new patterns, appearances, exploded views, interference check, surface analysis, OBJ and 3MF, toolbar customising and the Claude subscription option. Feature and tool names follow the software's new names.
2026-10-13	0.9.0-beta	First edition. Covers parts, sketching, gears, solid features, bodies, surfaces, reference geometry, the feature tree, sheet metal (including Convert to Sheet Metal with detected thickness and bend radii), drawings, evaluate tools, Ask Claude, troubleshooting and developer notes.

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